NATIONAL UNIVERSITY OF SINGAPORE

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CS1101S — PROGRAMMING METHODOLOGY

AY2023/2024 SEMESTER 1

READING ASSESSMENT 2

Time allowed: **40 minutes**

QUESTION PAPER

INSTRUCTIONS

1. This **Question Paper** contains **22 Questions**, and comprises **10 printed pages**, including this page.
2. You are also provided with an **Answer Sheet** to write your answers. It comprises **2 printed pages**.
3. Use a **2B pencil** to **write** your **Student Number** in the designated space on the front page of the **Answer Sheet**, and **shade** the corresponding circle **completely** in the grid for each digit or letter. DO NOT WRITE YOUR NAME.
4. You must **submit only** the **Answer Sheet** and no other documents.
5. All questions must be answered in the space provided in the **Answer Sheet**; no extra sheets will be accepted as answers.
6. For **multiple choice questions (MCQ)**, use a **2B pencil** to **shade** in the **circle** of the correct answer **completely**.
7. Each question has one correct answer. The **indicated marks** are awarded for each correct answer and there is **no penalty** for a wrong answer.
8. The **full score** of this assessment is **60 marks**.
9. Answer **all questions**.
10. This is a **closed-book** assessment.

(1) **[1 mark]** What is the **alphabet letter** that is in the box at the **top-right corner** on the **front page** of this **question paper**? (**Important**: Please make sure your answer is correct because it determines how your answers to all the subsequent questions will be graded.)

A. A
B. B
C. C
D. D

Section A [19 marks]

(2) **[3 marks]** What is the result of evaluating the following Source §3 program?

```
function f(g) {
    return g(x);
}
{
    const x = 1;
    f(y => y + 2);
}
```

A. Error: `Name x not declared.`
B. `undefined`
C. None of the other options is the correct answer
D. `2`
E. `3`
F. `1`
G. Error: `Name y not declared.`

(3) **[3 marks]** What is the result of evaluating the following Source §3 program?

```
let f = x => x + 1;

function g(x) {
    return y => f(x);
}
const h = g(2);
f = x => x + 3;
h(4);
```

A. `2`
B. None of the other options is the correct answer
C. `3`
D. `6`
E. `5`
F. `4`
G. `1`

(4) **[3 marks]** What is the result of evaluating the following Source §3 program?

```
let x = 1;

function f(y) {
    x = 2;
    let x = 8;
    return x + y;
}
f(4) + x;
```

A. 13
B. 12
C. 6
D. 8
E. 7
F. None of the other options is the correct answer
G. Error: `Cannot access 'x' before initialization.`
H. 14

(5) **[3 marks]** How many **function objects** get created during the evaluation of the following Source §3 program? (Do not count function objects of primitive and pre-declared functions.)

```
function f(x) {
    return x === 1
           ? y => 10
           : x === 2
           ? y => 20
           : x === 3
           ? y => 30
           : x === 4
           ? y => 40
           : y => 50;
}
f(1)(5);
```

A. More than 5
B. 3
C. 1
D. 4
E. 0
F. 5
G. 2

(6) **[4 marks]** What is the result of evaluating the following Source §3 program?

```
let f = x => {
    f = y => y + 2;
    return f(x) + 3;
};
f(1);
f(1);
```

A. Error: `Cannot assign new value to constant f.`

B. Error: `The function has encountered an infinite loop.`

C. `5`

D. None of the other options is the correct answer

E. `4`

F. `7`

G. `6`

H. `3`

(7) **[3 marks]** What is the result of evaluating the following Source §3 program?

```
let f = b => (f = b => !b)(!b);

while (f(true)) {
    f(true);
}
```

A. Error: `The loop has encountered an infinite loop.`

B. `true`

C. Error: `Cannot assign new value to constant f.`

D. `false`

E. None of the other options is the correct answer

F. `undefined`

G. `b => !b`

H. Error: `The function has encountered an infinite loop.`

Section B [22 marks]

For all the questions in this section, consider the following Source §3 program:

Program B:

```
// xs and ys must be lists of the same length
function fun(xs, ys) {
    if (is_null(xs)) {
        return null;
    } else {
        set_tail(ys, fun(tail(xs), tail(ys)));
        const new_pair = pair(head(xs), ys);
        return new_pair;
    }
}
const PP = list(list(1), list(2), list(3));
const QQ = list(4, 5, 6);
const RR = fun(PP, QQ);
```

(8) **[2 marks]** What is the value of `RR` (in *list notation*) at the end of the evaluation of Program B?

A. `list(list(1), 4, list(2), 5, list(3), 6)`
B. `list(1, 2, 3)`
C. `list(1, list(4), 2, list(5), 3, list(6))`
D. `list(list(1), 4, 5, 6)`
E. None of the other options is the correct answer
F. `list(1, 4, 2, 5, 3, 6)`
G. `list(list(1), list(4), list(2), list(5), list(3), list(6))`
H. `list(list(1), list(2), list(3))`

(9) **[2 marks]** What is the value of `PP` (in *list notation*) at the end of the evaluation of Program B?

A. `list(1, 2, 3)`
B. `list(1, 4, 2, 5, 3, 6)`
C. None of the other options is the correct answer
D. `list(list(1), 4, list(2), 5, list(3), 6)`
E. `list(1, list(4), 2, list(5), 3, list(6))`
F. `list(list(1), 4, 5, 6)`
G. `list(list(1), list(2), list(3))`
H. `list(list(1), list(4), list(2), list(5), list(3), list(6))`

(10) **[3 marks]** What is the value of `QQ` (in *list notation*) at the end of the evaluation of Program B?

A. `list(list(4), 2, list(5), 3, list(6))`
B. `list(4, list(2), 5, list(3), 6)`
C. `list(list(4), list(2), list(5), list(3), list(6))`
D. `list(list(4), list(5), list(6))`
E. `list(4, 2, 5, 3, 6)`
F. `list(4, 5, 6)`
G. None of the other options is the correct answer
H. `list(list(1), 4, list(2), 5, list(3), 6)`

(11) **[3 marks]** What is the result of the following statement if it is evaluated at the end of Program B?

```
[ PP === RR,  head(PP) === head(RR),  tail(PP) === tail(RR) ];
```

A. `[true,  true,  true ]`
B. `[true,  true,  false]`
C. `[true,  false, true ]`
D. `[true,  false, false]`
E. `[false, true,  true ]`
F. `[false, true,  false]`
G. `[false, false, true ]`
H. `[false, false, false]`

(12) **[3 marks]** How many **pairs** get created during the evaluation of Program B?

A. 24
B. 12
C. 15
D. 18
E. 6
F. 9
G. 3
H. None of the other options is the correct answer

(13) **[2 marks]** How many **bindings** does the **program environment frame** contain during the evaluation of Program B?

A. 2
B. 7
C. 4
D. 3
E. 6
F. More than 7
G. 1
H. 5

(14) **[4 marks]** How many **environment frames** get created during the evaluation of Program B? (Do not count the global environment frame. We assume that the applications of the primitive functions, such as `is_null`, `head`, `tail`, `pair`, `list`, `set_head`, and `set_tail`, do not create any frame.)

A. Fewer than 4
B. 7
C. 5
D. 6
E. 8
F. More than 9
G. 4
H. 9

(15) **[2 marks]** Of the **environment frames** that get created during the evaluation of Program B, how many **extend the program environment *directly***? (We assume that the applications of the primitive functions do not create any frame.)

A. 4
B. 0
C. 3
D. 6
E. 1
F. 5
G. 2
H. More than 6

(16) **[1 mark]** How many **function objects** get created during the evaluation of Program B? (Do not count function objects of primitive and pre-declared functions such as `array_length`, `math_floor`, `pair`, `head`, `tail`, `length`, and `map`.)

- **A.** 4
- **B.** More than 6
- **C.** 6
- **D.** 0
- **E.** 5
- **F.** 2
- **G.** 1
- **H.** 3

Section C [18 marks]

For all the questions in this section, consider the following Source §3 program:

Program C:

```
let A = [2, 1];

function check(A) {
    let i = 0;
    let sum = 0;

    while (i < array_length(A)) {
        let curr = A[i];
        sum = sum + curr;
        i = i + 1;
    }
    return () => sum > A[i - 1];
}

let curr = 0;

while (check(A)()) {
    A[array_length(A)] = curr;
    curr = curr - 1;
}

[curr, array_length(A)];
```

(17) **[4 marks]** What is the result of evaluating Program C?

A. `[-3, 6]`
B. `[-4, 7]`
C. `[-3, 7]`
D. `[-5, 7]`
E. `[-3, 5]`
F. `[-4, 6]`
G. None of the other options is the correct answer
H. `[-5, 8]`

(18) **[4 marks]** How many **environment frames** get created during the evaluation of Program C? (Do not count the global environment frame. We assume that the applications of the primitive functions, such as `array_length`, do not create any frame.)

A. 40
B. 16
C. 23
D. 31
E. More than 40
F. 14
G. 12
H. 9

(19) **[3 marks]** Of the **environment frames** that get created during the evaluation of Program C, how many **extend the program environment *directly***? (We assume that the applications of the primitive functions do not create any frame.)

A. 3
B. 9
C. 7
D. 6
E. 5
F. More than 9
G. 4
H. 8

(20) **[2 marks]** Of the **environment frames** that get created during the evaluation of Program C, how many extend directly from the environment frame in which the binding for the variable `i` has the final value 4?

A. 3
B. 5
C. 6
D. More than 8
E. 7
F. 2
G. 8
H. 4

(21) **[2 marks]** How many **bindings** does the **program environment frame** contain during the evaluation of Program C?

A. 1
B. More than 7
C. 6
D. 4
E. 5
F. 3
G. 2
H. 7

(22) **[3 marks]** How many **function objects** get created during the evaluation of Program C? (Do not count function objects of primitive and pre-declared functions.)

A. 2
B. 6
C. More than 9
D. 3
E. 7
F. 5
G. 1
H. 9

END OF QUESTIONS